Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech | **Time: 03 Hours**
Code: 155504 | **Subject: Cryptography and Network Security** | **Full Marks: 70**

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):- [2 x 7 = 14]

(a) How many distinct monoalphabetic substitution ciphers are possible over the English alphabet?
(i) 26
(ii) 2^{26}
(iii) 26^2
(iv) 26!

(b) Given the ciphertext "phhw ph diwhu wkh wrjd sduwb" encrypted using Caesar cipher, decrypt it using a key of 3.
(i) meet me after the toga party
(ii) kill me after the long march
(iii) join me before the town hall
(iv) stop me before the grand entry

(c) How many subkeys are generated in DES for its 16 rounds?
(i) 8
(ii) 10
(iii) 16
(iv) 32

(d) AES uses a 128-bit key and has 10 rounds. Calculate the total number of bytes used in all round keys (assuming one 16-byte key per round including the initial one).
(i) 128 bytes
(ii) 160 bytes
(iii) 176 bytes
(iv) 192 bytes

(e) Euler's totient function $\phi(91)$ equals:
(i) 82
(ii) 72
(iii) 90
(iv) 88

(f) Which of the following is NOT required in the ElGamal encryption system?
(i) A large prime p
(ii) A primitive root g of p
(iii) The totient $\phi(p)$
(iv) A random number k

(g) Which of the following best describes a digital signature?
(i) A public key for encryption
(ii) A hash encrypted with sender's private key
(iii) A secret key shared between users
(iv) A block cipher used to sign data

(h) X.509 certificates contain all of the following except:
(i) Symmetric session key
(ii) Certificate Authority's signature
(iii) Public key
(iv) Serial number

(i) Which of the following steps comes first in the TLS handshake process?
(i) Server sends certificate
(ii) Client sends encrypted key
(iii) Server verifies certificate
(iv) Data is transmitted

(j) In S MIME, the digital signature is created by:
(i) Encrypting the hash of the message with the recipient's public key
(ii) Encrypting the message with sender's private key
(iii) Hashing the message and encrypting the hash with the sender's private key
(iv) Hashing the message and sending it in plain form

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.2 (a) Compare passive and active attacks in computer security, providing examples for each. Explain briefly how appropriate security measures can mitigate these attacks. [7]

(b) What is a monoalphabetic substitution cipher? Explain how it works and illustrate with an example using a custom substitution key. Also, mention one advantage and one disadvantage. [7]

Q.3 (a) Explain the basic structure and working of the Data Encryption Standard (DES). [7]

(b) Explain the RC4 stream cipher algorithm with its key generation and encryption process. [7]

Q.4 (a) Compare stream ciphers and block ciphers. Give suitable examples and applications of each. [7]

(b) State and explain Fermat's Little Theorem. How is it used in cryptography? [7]

Q.5 (a) Apply the RSA algorithm. Given the prime numbers $p = 3$ and $q = 5$, and the public exponent $e = 5$, calculate the public key and private key, encrypt the message $M = 8$ using the public key, and decrypt the ciphertext using the private key. [7]

(b) Describe the Diffie-Hellman Key Exchange protocol with an example. [7]

Q.6 (a) Can data integrity and authentication be achieved solely by using a hash function? If yes, just draw the diagram of the solution. [3]

(b) Explain the process of digital signature. Why is it considered a bad idea to use the same symmetric key for both signing and verifying processes? [4]

(c) Differentiate between 'challenge-response' and 'Zero-knowledge' methods of authentication. Explain the step-by-step process of the Fiat- Shamir Protocol to achieve authentication. [7]

Q.7 (a) Describe the Kerberos authentication protocol in computer networks. Explain its components and authentication process. [10]

https://www.pyqonline.com

(b) What is non-repudiation? How digital signature provides non-repudiation? Explain with suitable diagram. [4]

© Apna BEU

Q.8 (a) What is Secure Socket Layer (SSL)? Explain the working of SSL protocol briefly. [7]

(b) How does HTTPS provide secure communication over the web? [7]

Q.9 (a) What is Pretty Good Privacy (PGP) and how does it provide secure email communication? Describe the role of public-key and symmetric encryption in PGP. [7]

(b) Describe the IP Security (IPSec) architecture. What are the roles of Authentication Header (AH) and Encapsulating Security Payload (ESP)? [7]

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play